Large language models identify causal genes in complex trait GWAS

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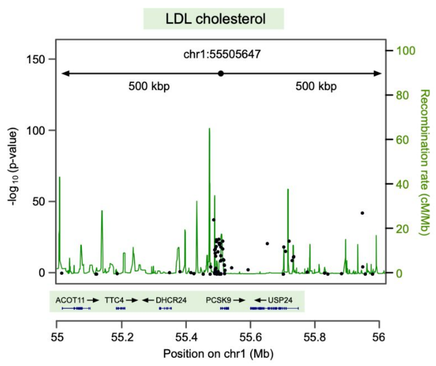
Background

Genome-wide association studies (GWAS) identify genomic regions with statistically significant genotype-phenotype associations.



Identifying the gene driving the association is challenging

Many statistical approaches, including literature mining, are used to identify likely causal genes.

Can large language models (LLMs) be used as a way of summarizing literature information to identify likely causal genes?

Methods

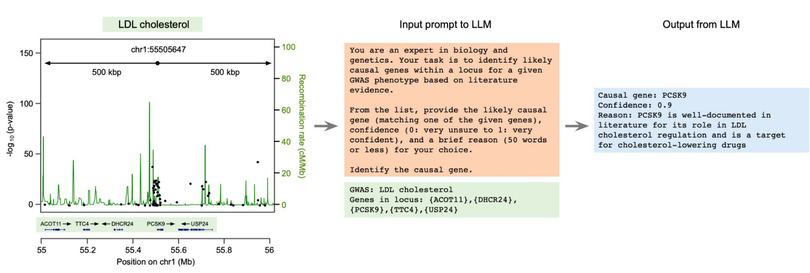


Figure 1. Schematic of the LLM-based approach for identifying causal genes at GWAS loci.

Evaluation datasets

We excluded common benchmark datasets like OpenTargets since they might be included in the LLM training data.

We used two datasets unlikely to be used in LLM training.

Dataset name	Number of loci	Ground truth source
GWAS catalog (post-April 2023)	641	Non-coding signals near coding signals
Weeks et al. (from authors)	1348	Non-coding credible sets near high-confidence coding credible set

Unpublished 23andMe Novel loci dataset

We use an unpublished dataset of 23 GWAS from 23andMe to test generalizability. We use a similar approach as Weeks et al. to assign ground-truth causal gene labels to loci.

A locus was defined as "novel" if its lead variant had a linkage disequilibrium (LD) $r^2 < 0.5$ with any variant in the GWAS catalog.

The final dataset has 403 loci.

Models evaluated

- OpenAI (gpt-3.5-turbo-0125, gpt-4-0613, gpt-4o-2024-08-06)
- Anthropic (Claude 3.5 Sonnet)
- Google (Gemini 1.5 Pro)
- Meta (Llama 3.1-405b)

Results

LLMs outperform existing methods on public benchmarks

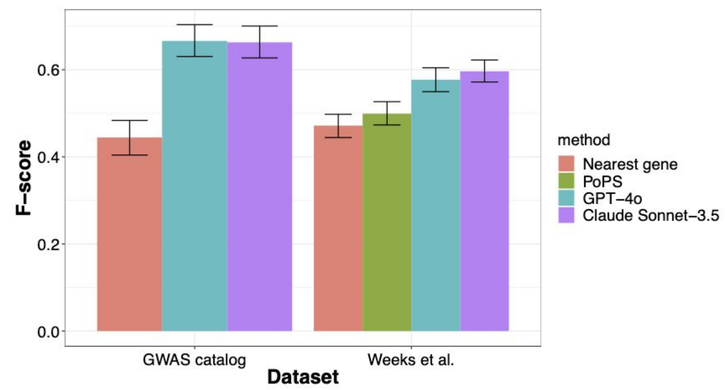


Figure 2. GPT-4o and Claude Sonnet-3.5 outperform "nearest gene" and PoPS on benchmark datasets.

LLMs generalize to novel loci and leverage semantic similarity

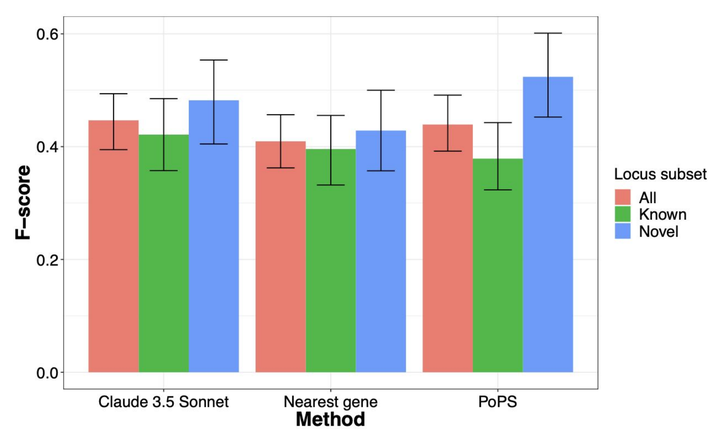


Figure 3. Performance of Claude Sonnet-3.5 is similar on known and novel loci.

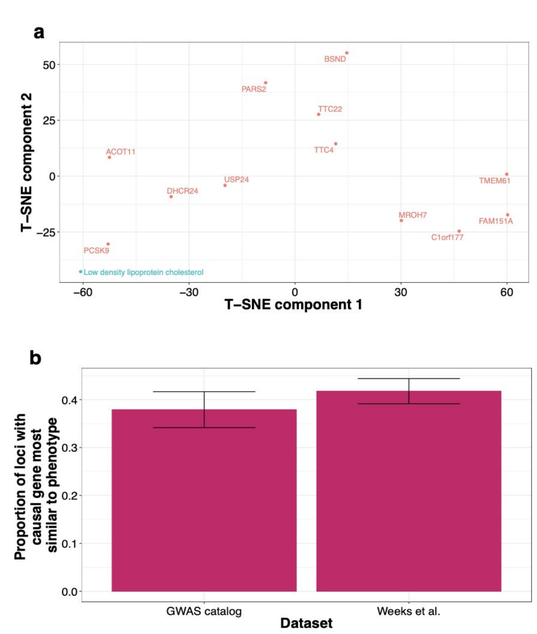


Figure 4. Embeddings from LLM-generated gene and phenotype summaries capture gene-phenotype similarity (a) at example locus (b) across benchmark datasets

Combining LLMs and existing methods gives best performance

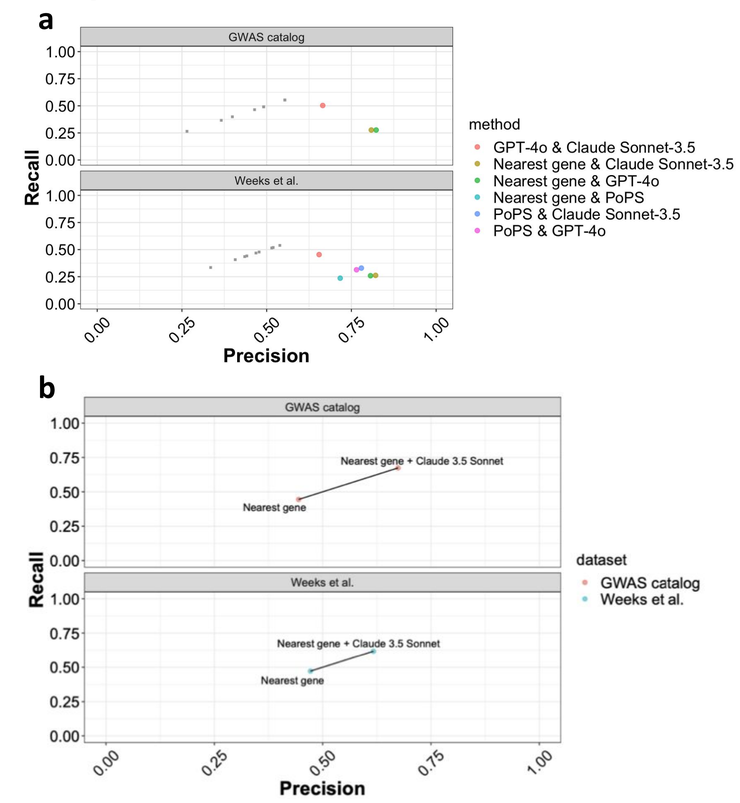


Figure 5. Combining LLMs and existing methods improves performance by (a) consensus (b) decision-tree ensemble.

LLM performance depends on literature information

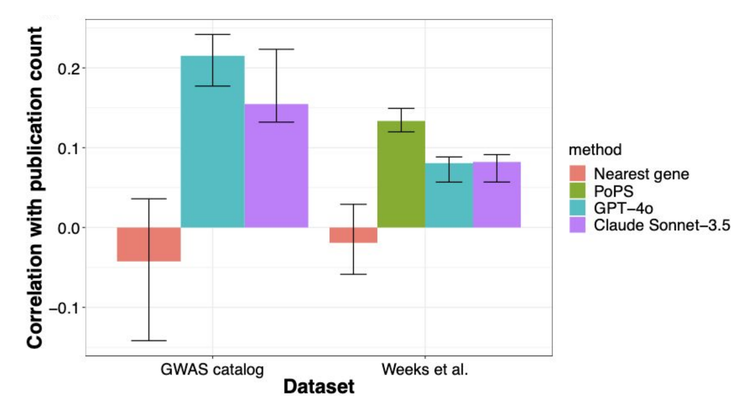


Figure 6. LLM prediction performance correlates positively with number of publications for causal gene.

Conclusions

- LLMs are a powerful, scalable, and cost-effective tool for causal gene prioritization.
- We show in a rigorous evaluation framework that LLMs outperform existing methods and perform similarly on known and novel loci.
- Combining LLMs and existing methods through ensembling significantly improves performance at causal gene prioritization.

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